

Is PPO inherently safe?

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1. Abstract

This paper investigates whether safe behavior emerges naturally in agents trained with Proximal Policy Optimization (PPO), or whether explicit constraint enforcement is necessary. We train a Brax Ant robot on a 2D navigation task under three constraint conditions: no constraint, soft constraint, and hard constraint. Each model is trained for 100M timesteps across 5 independent seeds under identical environment conditions. The obtained results show that PPO does not inherently learn safe behavior. The unconstrained agent accumulates unsafe behaviours despite extended training. Soft penalties marginally improve safety metrics but do not meaningfully deter unsafe behavior. Hard constraints enforce safety most effectively but at the cost of task performance. These findings suggest that safety must be explicitly designed into the reward and termination structure rather than expected to emerge from reward maximization alone.

2. Introduction & Background

Reinforcement learning is a branch of machine learning widely used for robotics applications. One of the most popular algorithms used is PPO, or Proximal Policy Optimization.

A PPO algorithm is a policy gradient method that optimizes a clipped surrogate objective to ensure stable policy updates for stable training. Agents trained with PPO can achieve very good performances across a wide range of control tasks, however performance and safety are not the same objective.

The literature on PPO mainly focuses on reward maximization and performance. Safety in its whole is overlooked, which is a problem since in safety-critical applications such as robotics, autonomous, and medical devices, an agent that maximizes reward while violating safety constraints is not deployable regardless of its task performance.

To implement the PPO algorithm, Brax was used because it is a fully differentiable rigid body physics engine implemented in JAX, designed for massively parallel simulation. The Ant morphology used in this work is a four-legged robot that must simultaneously learn locomotion and navigation, making it a non-trivial testbed for safety research.

3. Problem Formulation

The navigation task is defined in 2D. At the beginning of each episode, a goal position and three logical obstacles are sampled. The agent must navigate from its reset position to within a success radius of the goal while respecting safety constraints.

Safety constraints are defined as:

- **Obstacles collision**
- **Going out of bounds**
- **Speed violation**
- **Falling Over**

The three paradigms differ in how these constraints enter the reward and termination:

- **No constraint:** constraints are measured but do not affect reward or termination
- **Soft constraint:** constraint violations incur reward penalties but do not terminate the episode
- **Hard constraint:** constraint violations incur penalties and immediately terminate the episode

3.1 Reward Structure

The reward function encourages goal-directed behavior while discouraging inefficient or unsafe actions through a combination of positive reinforcement and penalties.

No Constraint: The agent receives a progress reward proportional to the reduction in distance to the goal, a success bonus upon reaching it, and a small action cost to promote energy-efficient movement. No safety penalties are applied.

Soft Constraint: Building on the baseline, the agent incurs penalties for collisions, out-of-bounds events, and falls, without episode termination. This creates a trade-off where the agent may still engage in unsafe actions if doing so yields higher cumulative reward.

Hard Constraint: Using the same reward components as the soft setting, any violation, but the speed violations, results in immediate episode termination. This acts as a strong implicit penalty by preventing the agent from accumulating future rewards after unsafe behavior, strongly incentivizing violation-free policies.

3.2 Termination Conditions

All three models share two common termination conditions: reaching the goal successfully and exceeding the maximum number of steps.

The soft-constraint model additionally penalizes collisions, out-of-bounds events, falls, and speed violations without terminating the episode. The hard-constraint model enforces the same penalties but goes further, triggering immediate episode termination on collisions, out-of-bounds events, and falls.

4. System Design

4.1 Environment

The environment is implemented using Brax, a differentiable physics engine designed for large-scale parallel simulation within the JAX framework. The agent is modeled as a quadrupedal Ant robot operating in a two-dimensional navigation setting.

At the beginning of each episode, the agent is initialized at a fixed starting position, while the goal location is randomly sampled from a predefined region of the environment. In addition, three obstacles are stochastically positioned between the start and goal locations.

The resulting task requires the agent to jointly learn locomotion and goal-directed navigation under spatial constraints. This combination introduces significant complexity, as the agent must simultaneously acquire stable control policies and develop obstacle-aware navigation strategies.

4.2 Observation Space

The observation space is represented by a 106-dimensional vector encompassing both the agent's proprioceptive state and key environmental information. This includes the standard Brax Ant observations, torso history, linear velocity, a goal vector, the relative XY positions of surrounding obstacles, and their respective radii.

The inclusion of obstacle-related features is critical, as it enables the agent to condition its policy on environmental structure and to learn anticipatory avoidance behaviors, rather than relying solely on reactive responses following unsafe interactions.

4.3 Domain Randomization

To enhance generalization and robustness, domain randomization is applied at the episode level. In particular, goal positions are sampled from a predefined distribution, and obstacle configurations vary across episodes. Additionally, minor perturbations may be applied to the agent's initial state.

This stochastic initialization prevents overfitting to specific environment configurations and encourages the learning of policies that generalize across a diverse set of scenarios. Consequently, the agent is better equipped to handle previously unseen configurations during evaluation.

5. Methods

5.1 PPO Implementation

We implement PPO from scratch in pure JAX. The actor and critic are two fully separate MLPs. Both networks take the raw enriched observation as input independently.

Baseline Metrics:

- **Success_bonus (+40.0):** Applied when reaching the goal.
- **Step penalty (−0.03):** Applied at every timestep.
- **Collision penalty (−5.0):** Applied when the agent collides with an obstacle, discouraging unsafe navigation.
- **Out-of-bounds penalty (−5.0):** Triggered when the agent leaves the valid area.
- **Speed penalty (−0.5):** Applied for undesirable speed behavior.
- **Fall penalty (−6.0):** Given when the agent falls.

Network architecture:

- Input layer: 106-dimensional observation vector(brax ant obs, torso history, velocity, goal vec, obstacle rel xy, obstacle radii)
- Hidden layer 1: 256 units, tanh activation
- Hidden layer 2: 256 units, tanh activation
- Actor output: 8-dimensional mean vector and a learned 8-dimensional log standard deviation
- Critic output: 1 scalar state-value estimate

PPO hyperparameters:

Parameter	Value	Parameter	Value
Learning rate	3e-4	Entropy coefficient	0.02
Discount factor γ	0.99	Max grad norm	1.0
GAE λ	0.95	PPO epochs	8
Clip ϵ	0.2	Steps per env	256
Value coefficient	0.5	Minibatch size	4096

5.2 Training Setup

- **Timesteps:** 100M per seed
- **Seeds:** 5 independent seeds (0, 1, 2, 3, 4)
- **Parallel environments:** 1024 via jax.vmap
- **Hardware:** 1 NVIDIA A100 80GB GPU per seed
- **Cluster:** Narval (Digital Research Alliance of Canada)
- **Total compute:** ~15 A100 GPU-days

5.3 Evaluation Protocol

Every 10 training iterations, the current policy is evaluated deterministically over 50 episodes. A final evaluation over 200 episodes is performed at the end of training. Metrics are averaged across seeds and reported as mean $\pm$ std.

6. Experiments and Results

6.1 Safety Violations & Task Performance

The no-constraint model exhibits a consistently high rate of safety violations throughout training, stabilizing at approximately 42.39 violations per 100 steps. This behavior indicates that, in the absence of explicit constraints, the agent does not acquire safe policies and instead prioritizes goal-reaching regardless of unsafe interactions, achieving a success rate of 29.9%.

The soft-constraint model demonstrates behavior closely aligned with the baseline, with violations remaining comparably high at 38.76 per 100 steps. This suggests that the imposed penalties are insufficient to effectively discourage unsafe actions, as the agent continues to tolerate violations in pursuit of the goal, yielding a similar success rate of 34.2%.

In contrast, the hard-constraint model achieves a substantial reduction in safety violations, dropping to just 7.04 per 100 steps. Since any violation results in immediate episode termination, the agent is strongly incentivized to adopt safer policies. However, this comes at the cost of task performance, as its success rate falls to 27.0%, reflecting the trade-off between safety and the ability to reach the goal.

6.2 Learning Curves

The no-constraint model demonstrates rapid convergence; however, it converges to policies that are inherently unsafe. The absence of constraints allows the agent to quickly exploit reward signals without regard for safety.

The soft-constraint model closely mirrors the learning dynamics of the baseline, further reinforcing the observation that our penalty-based approach alone is insufficient to meaningfully alter agent behavior in this setting.

The hard-constraint model exhibits slower learning progression. The enforcement of strict termination reduces effective exploration, thereby increasing the difficulty of discovering high-performing policies.

Additionally, variance across random seeds is notably higher in the hard-constraint setting, indicating increased sensitivity to initialization and stochasticity in the training process.

7. Discussion

7.1 Safety is Not Inherent in PPO

The experimental results demonstrate that Proximal Policy Optimization (PPO) does not inherently promote safe behavior. In the absence of explicit safety constraints, the agent consistently converges to policies that maximize reward while tolerating a high rate of safety violations. This suggests that PPO, as a general-purpose policy optimization algorithm, is fundamentally agnostic to safety considerations unless they are explicitly encoded in the

reward function or environment dynamics. Consequently, relying solely on standard reinforcement learning objectives is insufficient for safety-critical applications, where unsafe actions may yield high short-term rewards.

7.2 Hard Constraints Are Effective But Costly

Hard constraints emerge as the most effective mechanism for enforcing safety, as they directly prevent unsafe behaviors through immediate episode termination. This introduces a strong implicit penalty by eliminating the possibility of future reward accumulation following a violation. As a result, the agent is compelled to learn policies that strictly avoid unsafe states.

However, this effectiveness comes at a significant cost. Early termination reduces the effective exploration space, limiting the agent's ability to discover and refine successful trajectories. This leads to slower convergence and lower overall task performance. Furthermore, the increased variance across training runs suggests that hard constraints amplify sensitivity to stochastic factors, making training less stable.

7.3 The Safety-Performance Tradeoff

The results highlight a clear trade-off between safety and performance. While the no-constraint and soft-constraint models achieve higher success rates, they do so at the expense of safety, exhibiting frequent violations. In contrast, the hard-constraint model significantly reduces unsafe behavior but suffers from decreased task performance due to restricted exploration.

This trade-off underscores a fundamental challenge in reinforcement learning: optimizing for safety and performance simultaneously is non-trivial. Methods that strongly enforce safety may hinder learning efficiency, whereas methods that prioritize performance may fail to prevent unsafe behavior. Balancing these objectives remains an open problem and is critical for real-world deployment.

7.4 Reward Scale Matters

The findings also emphasize the importance of reward scaling in shaping agent behavior. In the soft-constraint setting, the penalty terms were insufficient to meaningfully influence the policy, as the agent continued to prioritize progress toward the goal despite incurring violations. This indicates that the relative magnitude of penalties compared to positive rewards plays a crucial role in determining whether safety constraints are effectively enforced.

If penalties are too small, they are effectively ignored; if they are too large, they may dominate the learning signal and hinder task performance as it causes the agent to just freeze. Therefore, careful tuning of reward components is essential to achieve a desirable balance between safety and efficiency.

8. Conclusion

This study investigated the emergence of safety in reinforcement learning using PPO under different constraint formulations. The results confirm that safety is not an inherent property of the learning process, as unconstrained agents consistently converge to unsafe policies that prioritize reward maximization.

Soft constraints prove ineffective, as agents continue to tolerate violations when they yield higher cumulative return. Hard constraints, enforced through episode termination, significantly reduce unsafe behavior but introduce a trade-off between safety and task performance, resulting in slower learning and lower success rates.

Overall, these findings highlight the need for more principled approaches to safety in reinforcement learning, particularly in applications where reliability and robustness are critical.

9. Future Work

Future research can build upon this work by exploring more advanced approaches to balancing safety and performance in reinforcement learning.

One promising direction is the use of smaller environment steps combined with stronger or adaptive penalty scaling mechanisms (Lagrangian method), where the magnitude of penalties evolves during training to better influence agent behavior. Additionally, constrained reinforcement learning frameworks, such as Constrained Policy Optimization (CPO), could provide a more principled way to enforce safety constraints while preserving performance, as it directly optimizes within a feasible region defined by safety thresholds rather than relying on penalty terms alone.

Another avenue involves the integration of risk-sensitive objectives, which explicitly account for uncertainty and worst-case outcomes rather than optimizing expected return alone. Improvements in observation design may also enhance the agent's ability to anticipate and avoid obstacles, leading to more robust policies.

Finally, extending this work to more complex environments or real-world robotic systems would provide valuable insights into the scalability and practical applicability of these approaches in safety-critical domains.

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